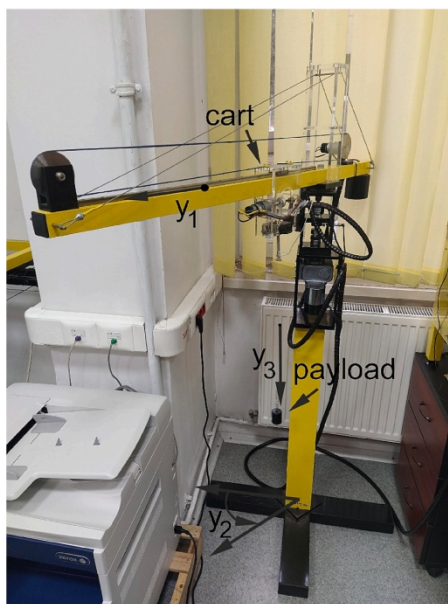




A. Laboratory tower-crane safe control

EVL L2 | AM1



(a) Lifting of the concrete block



(b) Obstacle avoidance



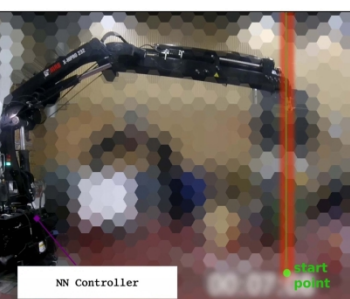
(c) Transporting of the concrete block



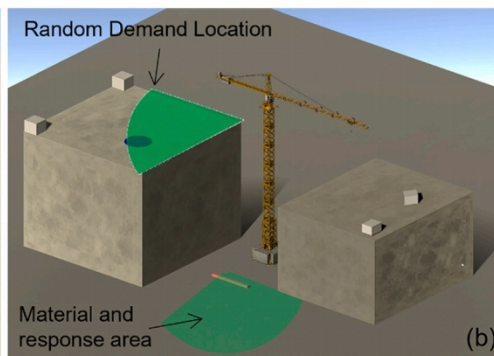
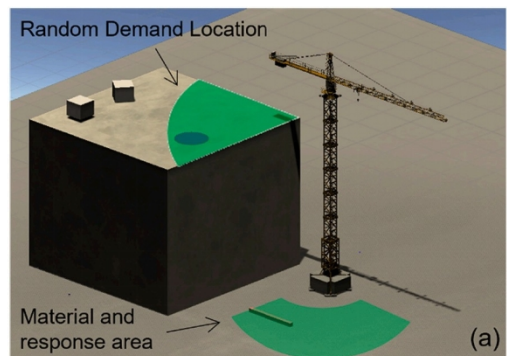
(d) Reach the target position

B. Robotized mobile-crane placement and assistance control system

(i) EVL L3 | AM2

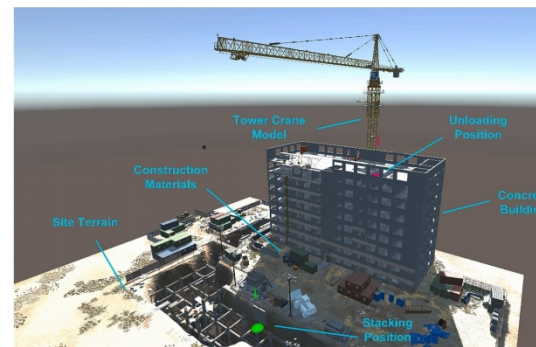


(ii) EVL L2 | AM1

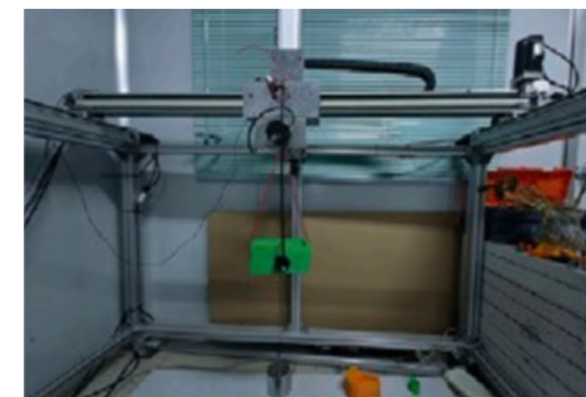


C. Tower-crane lift-path planning

(i) EVL L2 | AM0



(ii) EVL L0 | AM0



D. Double-pendulum crane trajectory planning

EVL L2 | AM1

EVL = validation exposure; **AM** = runtime assurance